

Programmed cell death (PCD) is involved in both cellular development and aging. Apoptosis is a form of PCD that generally does not induce a strong immune response. However, during another PCD mechanism called necroptosis, dying cells release molecules that promote an inflammatory immune response that can cause tissue damage.

In mammals, necroptotic pathways are mediated by the RIPK3 kinase. After phosphorylation by another kinase, RIPK3 phosphorylates the MLKL protein, which disrupts plasma membrane structure and function. Scientists suspected that necroptosis is involved in the natural aging of the male reproductive system. In sexually mature wild-type (WT) mice of all ages, RIPK3 expression was confirmed in testicular seminiferous tubule cells (spermatogonia and Sertoli cells) but not in testicular Leydig cells. In an experiment, the scientists knocked out the *RIPK3* gene in male mice and monitored their reproductive systems over time.

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Table 2 shows mean seminal vesicle masses and serum testosterone levels for equally sized groups of 18-month-old WT and *RIPK3*^{-/-} mice.

Table 2 Seminal Vesicle Mass and Serum Testosterone Levels in 18-Month-Old WT and *RIPK3*^{-/-} Mice with Standard Error of the Mean

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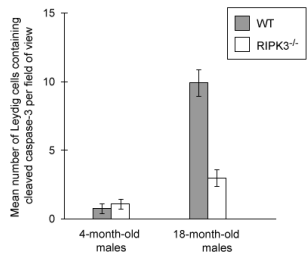


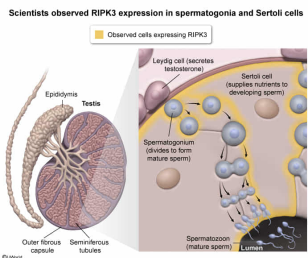
Figure 1 Presence of cleaved caspase-3 in Leydig cells of WT and *RIPK3*^{-/-} mice. Error bars represent the 95% confidence interval.

In the experiments described in the passage, the scientists observed *RIPK3* expression in cells that:

- I. release testosterone into the bloodstream.
 - II. mitotically and meiotically divide into cells that ultimately become mature sperm.
 - III. supply nutrients to sperm cells undergoing development.
 - IV. produce seminal fluid.
- ☐ A. I and II only [7%]
☒ B. II and III only [36%]
☐ C. I, II, and III only [16%]
☐ D. II, III, and IV only [39%]

Correct
37% answered correctly

Explanation:



Gonads (ie, male **testes** and female **ovaries**) are reproductive glands that produce **gametes** (ie, eggs and sperm). In males, the testes contain specialized, coiled structures called **seminiferous tubules**, which are the locations of **spermatogenesis** (ie, sperm production). Within the seminiferous tubules, spermatogenesis **proceeds** as follows:

- Spermatogonia** undergo **mitotic division**, forming two daughter cells. One of these daughter cells remains a spermatogonium while the other becomes a **primary spermatocyte** and begins **meiosis**. Meiosis I results in two **secondary spermatocytes**.
- Completion of meiosis II by the **secondary spermatocytes** yields four **spermatids**.

3. The spermatids undergo several changes (eg, cytoplasm loss, acrosome formation, tail development), ultimately developing into [spermatozoa](#).
4. The spermatozoa migrate to the epididymis where they continue maturation and acquire motility.

The seminiferous tubules also contain specialized "nurse cells" called **Sertoli cells** that are responsible for providing nutrients to sperm cells as well as regulating cell development. [Testosterone](#), a hormone that stimulates sperm cell differentiation, is secreted by **Leydig cells**, or endocrine cells found in interstitial tissue surrounding the seminiferous tubules.

In the passage, scientists examined the expression of RIPK3 (a [kinase](#) involved in necroptotic [programmed cell death](#)) in testicular cells from sexually mature WT mice of all ages. **RIPK3 expression** was observed in **spermatogonia**, which **mitotically and meiotically divide into cells that ultimately become mature sperm**, and **Sertoli cells**, which **supply nutrients to sperm cells undergoing development (Numbers II and III)**.

(Number I) Leydig cells release testosterone; however, the passage states that RIPK3 expression was *not* confirmed in Leydig cells.

(Number IV) Three accessory glands are responsible for seminal fluid production. The seminal glands produce the largest portion of seminal fluid whereas the prostate gland produces fluid containing enzymes necessary to prevent sperm coagulation within the vagina. The bulbourethral (Cowper's) glands produce a thick, alkaline mucus that lubricates the penis tip and protects sperm from the acidic environment in the urethra during ejaculation. Cells comprising these accessory glands were *not* tested for RIPK3 expression.

Educational objective:

During spermatogenesis in the seminiferous tubules, spermatogonia undergo mitosis, forming one daughter cell that remains a spermatogonium and one daughter cell that becomes a primary spermatocyte. Primary spermatocytes undergo meiosis, ultimately forming spermatids that develop into spermatozoa (mature sperm). In addition, Sertoli cells supply nutrients to sperm cells during spermatogenesis.

Time spent: 0 sec
Subject:
Biology

QID: 403453
System:
Reproduction

Programmed cell death (PCD) is involved in both cellular development and aging. Apoptosis is a form of PCD that generally does not induce a strong immune response. However, during another PCD mechanism called necroptosis, dying cells release molecules that promote an inflammatory immune response that can cause tissue damage.

In mammals, necroptotic pathways are mediated by the RIPK3 kinase. After phosphorylation by another kinase, RIPK3 phosphorylates the MLKL protein, which disrupts plasma membrane structure and function. Scientists suspected that necroptosis is involved in the natural aging of the male reproductive system. In sexually mature wild-type (WT) mice of all ages, RIPK3 expression was confirmed in testicular seminiferous tubule cells (spermatogonia and Sertoli cells) but not in testicular Leydig cells. In an experiment, the scientists knocked out the *RIPK3* gene in male mice and monitored their reproductive systems over time.

Table 1 shows fertility rates of *RIPK3*-knockout mice (*RIPK3*^{-/-}) of different ages and age-matched WT mice. To test fertility, all males were mated with WT females of peak reproductive age (about 10 weeks old).

Table 1 Effect of *RIPK3* Knockout on Fertility

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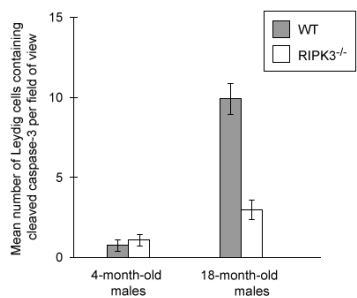


Figure 1 Presence of cleaved caspase-3 in Leydig cells of WT and *RIPK3*^{-/-} mice. Error bars represent the 95% confidence interval.

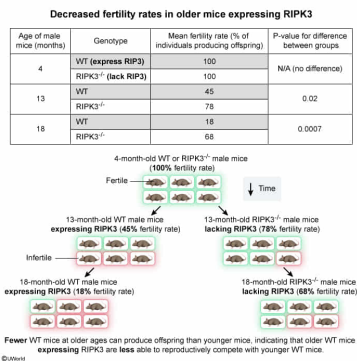
Which of the following conclusions is best supported by the data presented in Table 1?

- ☐ A. *RIPK3*-induced inflammation most significantly affects reproductive system function in 13-month-old male mice. [4%]
- ☐ B. Knockout of *RIPK3* allows more male mice to produce offspring at older ages than at younger ages due to reduced necroptosis of reproductive structures. [33%]
- ☒ C. *RIPK3* expression significantly reduces the ability of older male mice in a population to reproductively compete with younger male mice. [44%]
- ☐ D. Knockout of *RIPK3* prevents PCD from occurring in the reproductive systems of 13-month-old and 18-month-old male mice. [17%]

Correct

45% answered correctly

Explanation:



All organisms eventually undergo **senescence**, a natural aging process in which cells and tissues acquire damage (eg, **mutations**) and diminish in function over time.

In cells, senescence occurs when **cell cycle** progression can no longer occur. Cellular senescence may result from degradation of chromosome ends (ie, **telomeres**) over time. Specifically, chromosomes degrade with each of round of the cell cycle because **DNA replication** processes **do not allow** full replication of linear chromosome ends.

Some senescent cells may be eliminated via **programmed cell death** (PCD), but effects of senescence accumulate over time, eventually decreasing organism function (eg, reproduction).

In the passage, scientists suspected that **necroptosis**, a RIPK3-mediated PCD pathway, is involved in the natural aging of the male reproductive system. These scientists experimentally knocked out the *RIPK3* gene in male mice and monitored their reproductive systems over time.

Table 1 shows that WT mice (ie, those **expressing** RIPK3) exhibited **greatly reduced** fertility rates at older ages (13 and 18 months) than at younger ages (4 months) when mated with females. However, knocking out the *RIPK3* gene (ie, preventing RIPK3 expression), **significantly decreased** fertility loss in aging mice. In other words, **more** $RIPK3^{-/-}$ mice (ie, those **lacking** RIPK3) were able to continue producing offspring at older ages than WT mice **expressing** RIPK3 (***p*-value < 0.05**).

Older mice expressing RIPK3 (13-month-old and 18-month-old WT mice) **exhibited much lower fertility rates** (ie, lower percentage of individuals producing offspring) than younger 4-month-old mice. Therefore, RIPK3 expression likely **significantly reduces** the ability of **older** mice to **reproductively compete** with younger mice.

(Choice A) The data in Table 1 show that RIPK3 expression **decreases** fertility rates by a **higher percentage** (and with a **more** significant *p*-value) in 18-month-old mice than in 13-month-old-mice.

(Choices B and D) $RIPK3^{-/-}$ mice exhibited **higher** fertility rates at older ages than **age-matched WT** mice. However, a **lower** (**not** higher) percentage of $RIPK3^{-/-}$ mice were able to produce offspring at older ages (13 and 18 months) than at younger ages (4 months). This result indicates that **some** PCD and tissue damage likely still occurs in the reproductive systems of older $RIPK3^{-/-}$ males.

Educational objective:

Senescence is a natural aging process in which cells and tissues acquire damage and diminish in function over time. Programmed cell death (PCD) may eliminate some senescent cells from organisms; however, accumulating effects of senescence over time cause organism functions to decline.

Time spent:0 sec
Subject:
Biology

QID:403454
System:
Reproduction

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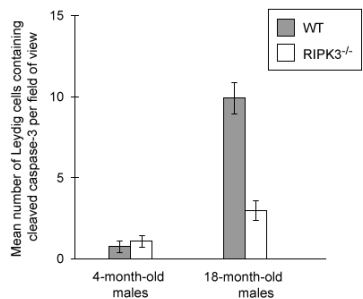


Figure 1 Presence of cleaved caspase-3 in Leydig cells of WT and *RIPK3*^{-/-} mice. Error bars represent the 95% confidence interval.

Based on Table 2, which group(s) likely exhibit more variation around the mean values in seminal vesicle mass and serum testosterone levels?

- ☐ A. WT only [4%]
- ☐ B. *RIPK3*^{-/-} only [3%]
- ☐ C. *RIPK3*^{-/-} and WT, respectively [25%]
- ☒ D. WT and *RIPK3*^{-/-}, respectively [66%]

Correct
65% answered correctly

Explanation:

Calculating range using standard error of the mean (SEM) from Table 2

Genotype	Mean seminal vesicle mass (mg)	Mean serum testosterone level (pg/mL)
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Minimum WT mean = 2,900 - 500 = 2,400 mg

Maximum WT mean = 2,900 + 500 = 3,400 mg

WT mean range = 2,400 to 3,400 mg

Minimum *RIPK3*^{-/-} mean = 542 - 400 = 142 mg

Maximum *RIPK3*^{-/-} mean = 542 + 400 = 942 mg

RIPK3^{-/-} mean range = 142 to 942 mg

Minimum WT mean = 375 - 150 = 225 pg/mL

Maximum WT mean = 375 + 150 = 525 pg/mL

WT mean range = 225 to 525 pg/mL

Minimum *RIPK3*^{-/-} mean = 1,010 - 250 = 760 pg/mL

Maximum *RIPK3*^{-/-} mean = 1,010 + 250 = 1,260 pg/mL

RIPK3^{-/-} mean range = 760 to 1,260 pg/mL

For equal-sized groups, larger range indicates greater variation

The **standard error of the mean (SEM)** is a statistical measure that provides information on the *precision* of a measurement. Specifically, SEM gives information on how close the *calculated mean* value for an experimental group is to the *true* mean value for the population, with a *smaller* SEM indicating that the calculated mean is *closer* to the true population mean. SEM depends on both the **standard deviation (SD)** for an experimental group as well as the sample size.

SD refers to the amount by which individual data points *vary* around the group mean. As SD (ie, data *variability*) *increases*, SEM also *increases*. In addition, *increasing* the size of a sample will *decrease* SEM.

In the passage, Table 2 shows mean seminal vesicle mass (mg) and mean serum testosterone level (pg/mL), along with SEM values, for equally sized groups of WT mice (ie, mice expressing RIPK3) and RIPK3^{-/-} mice (ie, mice lacking RIPK3).

For mean **seminal vesicle mass**, WT mice exhibited a **higher SEM (± 500 mg)** than the RIPK3^{-/-} mice (± 400 mg), indicating that WT mice likely have the greater variation around the mean value for this measurement. In contrast, for mean **serum testosterone level**, RIPK3^{-/-} mice exhibited a **higher SEM (± 250 pg/mL)** than the WT mice (± 150 pg/mL). Therefore, RIPK3^{-/-} mice likely have more variation around the mean for this measurement. Consequently, the groups that likely exhibit **more variation** around the mean values in seminal vesicle mass and serum testosterone levels are **WT mice and RIPK3^{-/-} mice, respectively (Choices A, B, and C)**.

Educational objective:

The standard error of the mean (SEM) is indicative of how close the calculated mean value for an experimental group is to the true mean value for the population. As standard deviation (ie, variation among data points) increases, SEM also increases.

Time spent:0 sec	QID:403455
Subject:	System:
Biology	Reproduction

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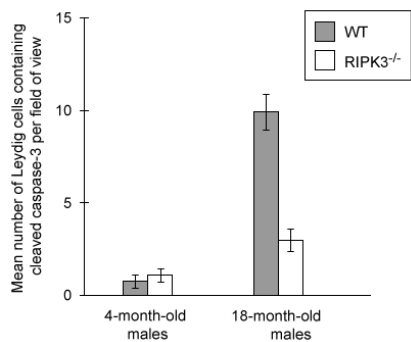


Figure 1 Presence of cleaved caspase-3 in Leydig cells of WT and *RIPK3*^{-/-} mice. Error bars represent the 95% confidence interval.

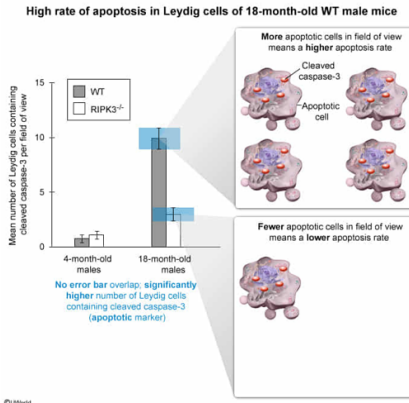
The data shown in Figure 1 indicate that Leydig cells:

- ☐ A. underwent less apoptosis in 4-month-old WT mice than in 4-month-old RIPK3^{-/-} mice. [5%]
- ☐ B. transcribed the caspase-3 gene most rapidly in 18-month-old WT mice. [19%]
- ☐ C. activated only nonapoptotic PCD pathways in 4-month-old RIPK3^{-/-} mice. [3%]
- ✓ ☒ D. exhibited the highest rates of apoptotic PCD in 18-month-old WT mice. [71%]

Correct

70% answered correctly

Explanation:



Programmed cell death (PCD) via **apoptosis** plays a key role in several process, including **embryonic development** (eg, sculpting of the **hand**) and the removal of damaged, infected, or senescent cells (ie, cells that can no longer divide via the **cell cycle**).

In the passage, scientists tested **Leydig cells** from 4-month-old and 18-month-old WT and RIPK3^{-/-} mice for the presence of a known marker of apoptosis (ie, cleaved caspase-3). Leydig cells stained for cleaved caspase-3 were microscopically visualized and the mean number of cells per field of view containing cleaved caspase-3 was determined for each group (Figure 1).

Figure 1 shows that 18-month-old WT mice had the **greatest** number of Leydig cells containing cleaved caspase-3. In addition, the 95% confidence interval **error bar** for this data point does *not* overlap with any of the other error bars, indicating that this measurement is statistically **significant**. Therefore, because cleaved caspase-3 is a marker of apoptosis, Leydig cells in 18-month-old WT mice likely exhibited the **highest rates of apoptotic PCD** compared to the other groups.

(Choice A) Although Figure 1 shows that the mean number of Leydig cells containing cleaved caspase-3 was lower in 4-month-old WT mice than in 4-month-old RIPK3^{-/-} mice, the error bars for these two groups **overlap**. This indicates that the difference in apoptosis levels between these two groups was *not* statistically significant.

(Choice B) Although 18-month-old WT mice had the *most* Leydig cells containing cleaved caspase-3 (ie, a protein), Figure 1 does *not* provide information on the rate of **transcription** (ie, rate of mRNA production) for the gene encoding caspase-3.

(Choice C) Figure 1 shows that very *low* numbers of Leydig cells from 4-month-old RIPK3^{-/-} mice contained cleaved caspase-3, indicating that apoptosis was likely occurring at *low* levels in the Leydig cells of these mice. However, Figure 1 only provides information on the presence of this *apoptotic* marker, *not* on how nonapoptotic PCD pathways may impact Leydig cells.

Educational objective:

Programmed cell death via apoptosis is involved in many processes, including embryonic development and the removal of damaged, infected, or senescent cells (ie, cells that can no longer divide via the cell cycle).

Time spent: 0 sec
Subject:
Biology

QID: 403456
System:
Reproduction